

Highlights

Explicit Second-Order Bounds on the Domain of Validity for Lyapunov–Schmidt Reduction

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- Derive explicit second-order validity bounds for Lyapunov–Schmidt reduction.
- Admissible radii admit an explicit closed-form geometric characterisation.
- Bounds evaluated in closed form for Hopfield networks on connected regular graphs.
- Second-order bounds proved contained in published first-order bounds for regular graphs.
- Simple counterexample shows that neither bound family dominates the other in general.

Explicit Second-Order Bounds on the Domain of Validity for Lyapunov–Schmidt Reduction

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ABSTRACT

Lyapunov–Schmidt reduction is a widely used dimensionality reduction technique for bifurcation analysis in high-dimensional systems. While classical formulations guarantee the local existence of reduced-order equations, they typically lack explicit quantitative estimates on the size of the neighbourhoods in which these representations faithfully capture the full bifurcation structure. In recent work, we have addressed this limitation by deriving bounds on the domain of validity of this reduction using first-order conditions on the vector field together with a quantitative result for the implicit function theorem. In this article we explore beyond, and adopt second-order conditions that incorporate the Hessians of the vector field to develop these bounds, and obtain a new set of results. We then investigate the applicability of these newly derived bounds to Hopfield-like networked dynamical systems, evaluate these bounds for pitchfork bifurcations on connected regular graphs, and finally examine the relationship between the two certified domains for this class of systems.

1. Introduction

Local bifurcations are points of transition in the qualitative behaviour of nonlinear dynamical systems, encompassing fundamental changes in the nature of equilibria - stability and count, emergence of limit cycles, and multistability - arising as system parameters cross critical thresholds [11]. Bifurcation analysis plays a central role in the design and regulation of such behaviours across a wide range of applied systems, including ecology [25, 4], climate dynamics [1, 19], cell biology [28, 21], computational neuroscience [16], cardiac dynamics [24], and transportation [22].

As the dimensionality of a dynamical system grows, analytical treatment of the vector fields involved grows increasingly difficult. A key step in rendering bifurcation analysis of high-dimensional systems tractable is dimensionality reduction, since bifurcations generically occur on low-dimensional manifolds [23, 18]. The Lyapunov–Schmidt reduction is a widely used tool for this purpose [9, 26], reducing the problem to a lower-dimensional algebraic system whose zeros correspond one-to-one with equilibria of the full system near the bifurcation point. This construction is, however, intrinsically local: the size of the neighbourhood over which the reduced equations faithfully represent the full bifurcation diagram is typically unknown. Explicit bounds addressing this were first presented in [13], building on Implicit Function theorem domain estimates from [17], and specialised in [12] to Hopfield-type networked dynamical systems, evaluated explicitly for connected regular graphs.

The previously derived first-order bounds use one of two formulations of the Implicit Function theorem validity bounds in [17]; the alternative instead incorporates Hessians

alongside Jacobians, giving a validity condition with a different computational structure. Because the two formulations follow distinct proof strategies, neither is guaranteed a priori to certify the larger neighbourhood, and we regard the resulting bounds as complementary rather than successive refinements of one another. This second-order structure further offers gains the first-order bounds lack: an admissible neighbourhood determined by explicit algebraic conditions that are numerically simpler and geometrically characterisable, and which may themselves be relaxed to readily computable conservative estimates without forfeiting the validity guarantee. This article develops these *second-order* bounds for the general Lyapunov–Schmidt reduction setting, specialises them to the same networked systems considered previously, and shows that, for this class, the resulting domain is contained within the first-order bound. We further show that this inclusion need not hold in general, that the regular-graph result is a special case of a broader degeneracy condition, and finally contrast the computational and geometric structure of the two bounds families.

The remainder of this paper is organised as follows. Section 2 establishes the necessary notation and recalls the foundational prior results on explicit second-order domain estimates for the Implicit Function theorem and the formulation of the Lyapunov–Schmidt reduction. Section 3 presents the main contribution: explicit second-order bounds on the validity domain of the Lyapunov–Schmidt reduction and characterises the resulting admissible region. Section 4 applies these bounds to Hopfield-like networked systems, and evaluates them analytically at pitchfork bifurcations on connected regular graphs. Section 5 provides a quantitative and qualitative comparison of the second-order bounds with the first-order bounds established in prior work. Section 6 presents numerical studies that illustrate the theoretical results, and supporting derivations are collected in Appendices A and B.

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2. Background

2.1. Notation and Definitions

For $z \in \mathbb{R}^n$, we use $\mathcal{N}(z) \subset \mathbb{R}^n$ to denote an open neighbourhood of z . For $z_0 \in \mathbb{R}^n$ and $r > 0$, the open ball of radius r centred at z_0 is given by $\mathcal{B}(z_0, r) := \{z \in \mathbb{R}^n : \|z - z_0\|_2 < r\}$, where $\|\cdot\|_2$ is the Euclidean 2-norm on $\mathbb{R}^n$. We denote the spectral and Frobenius norms of an arbitrary tensor A by $\|A\|_2$ and $\|A\|_F$ respectively. For a given rank-3 tensor $T \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and rank-2 tensors $U \in \mathbb{R}^{n_2 \times m_2}$ and $V \in \mathbb{R}^{n_3 \times m_3}$, let $T[U, V]$ denote the bilinear contraction of T with U and V , defined as $[T[U, V]]_{ijk} := \sum_{\mu, \nu} T_{i\mu\nu} U_{\mu j} V_{\nu k}$. We write $\mathbf{0}_n, \mathbf{1}_n \in \mathbb{R}^n$ for the all-zero and all-one column vectors respectively, and $\mathbf{0}_{p \times q}, \mathbf{1}_{p \times q} \in \mathbb{R}^{p \times q}$ for their matrix versions. The order n identity matrix is denoted by $I_n \in \mathbb{R}^{n \times n}$. For a rank-3 tensor $T \in \mathbb{R}^{l \times (m_1 + m_2) \times (n_1 + n_2)}$, the notation

$$T = \begin{bmatrix} A & B \\ C & D \end{bmatrix}_i$$

denotes a block representation of T , where the j and k index ranges are partitioned into row and column blocks $A \in \mathbb{R}^{l \times m_1 \times n_1}, B \in \mathbb{R}^{l \times m_1 \times n_2}, C \in \mathbb{R}^{l \times m_2 \times n_1}, D \in \mathbb{R}^{l \times m_2 \times n_2}$, while the i -index is shared by all blocks.

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph with vertex set $\mathcal{V}$ of n nodes, and edge set $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$. We associate with $\mathcal{G}$ the signed adjacency matrix $A \in \mathbb{R}^{n \times n}$, where a_{ij} is the signed weight of the edge between nodes i and j , and $a_{ij} = 0$ if no such edge exists. A graph is k -regular if every vertex has degree k . Equivalently, the binary adjacency matrix A (unit weight on existing edges) of a k -regular graph satisfies $A\mathbf{1}_n = k\mathbf{1}_n$.

2.2. Implicit Function theorem

Theorem 2.1. [30, Theorem 4.B] Let $f : U \times V \rightarrow \mathbb{R}^m$ be a smooth map, where $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ are open sets. Consider the system of equations $f(u, v) = f(u_0, v_0)$ for some $(u_0, v_0) \in U \times V$ for which $D_v f(u_0, v_0)$ is invertible. Then, there exist open sets $\mathcal{N}(u_0) \subset U$ and $\mathcal{N}(v_0) \subset V$ equipped with a unique mapping $g : \mathcal{N}(u_0) \rightarrow \mathcal{N}(v_0)$ such that $g(u_0) = v_0$ and $f(u, g(u)) = f(u_0, v_0)$, $\forall u \in \mathcal{N}(u_0)$.

Theorem 2.2. Restating [17, Theorem 3.3], we define the following quantities:

$$M_u := \|D_u f(u_0, v_0)\|, \quad M_v := \|[D_v f(u_0, v_0)]^{-1}\|. \quad (2.1)$$

For given $R_u, R_v > 0$ such that $\mathcal{B}(u_0, R_u) \subset U$ and $\mathcal{B}(v_0, R_v) \subset V$, let $\Omega := \mathcal{B}(u_0, R_u) \times \mathcal{B}(v_0, R_v)$. Define

$$K_{uu} := \sup \left\{ \|D_u^2 f(u, v)\| : (u, v) \in \Omega \right\}, \quad (2.2a)$$

$$K_{uv} := \sup \left\{ \|D_u D_v f(u, v)\| : (u, v) \in \Omega \right\}, \quad (2.2b)$$

$$K_{vv} := \sup \left\{ \|D_v^2 f(u, v)\| : (u, v) \in \Omega \right\}. \quad (2.2c)$$

Then for all $0 < r_u < R_u$ and $0 < r_v < R_v$ satisfying

$$\frac{1}{2} K_{uu} r_u^2 + K_{uv} r_u r_v + \frac{1}{2} K_{vv} r_v^2 < \frac{r_v}{M_v} - M_u r_u, \quad (2.3a)$$

$$K_{uv} r_u + K_{vv} r_v < \frac{1}{M_v}, \quad (2.3b)$$

simultaneously, the implicit map g from Theorem 2.1 is $\mathcal{C}^2$ and is valid over $g : \mathcal{B}(u_0, r_u) \rightarrow \mathcal{B}(v_0, r_v)$.

2.3. Lyapunov–Schmidt reduction

Let $X \subset \mathbb{R}^n$ and $\Lambda \subset \mathbb{R}^m$ be open sets. Consider the parametrised nonlinear dynamical system $\dot{x} = \Phi(x, p)$ where $x \in X$ is the state vector, $p \in \Lambda$ is the parameter vector, and $\Phi : X \times \Lambda \rightarrow \mathbb{R}^n$ is at least $\mathcal{C}^2$, with a bifurcation of equilibria at some singular point $(x_*, p_*) \in X \times \Lambda$, i.e., $\Phi(x_*, p_*) = 0$ and $J_* := D_x \Phi(x_*, p_*)$ is singular with $q := \dim(\ker J_*) > 0$. We write the SVD of J_* as:

$$J_* = [W_* \quad \overline{W}_*] \begin{bmatrix} \Sigma_* & \mathbf{0}_{(n-q) \times q} \\ \mathbf{0}_{q \times (n-q)} & \mathbf{0}_{q \times q} \end{bmatrix} \begin{bmatrix} \overline{V}_*^\top \\ V_*^\top \end{bmatrix} = W_* \Sigma_* \overline{V}_*^\top, \quad (2.4)$$

where $\Sigma_* = \text{diag}(\sigma_1, \dots, \sigma_{n-q})$ with $\sigma_1 \geq \dots \geq \sigma_{n-q} > 0$ is the diagonal matrix of the nonzero singular values of J_* , and $q < n$. Here, $V_*, \overline{W}_* \in \mathbb{R}^{n \times q}$ correspond to the null singular values and $\overline{V}_*, W_* \in \mathbb{R}^{n \times (n-q)}$ correspond to the nonzero singular values. By the properties of the SVD, the columns of V_* and W_* form orthonormal bases for the null and range spaces of J_* , while those of $\overline{V}_*$ and $\overline{W}_*$ form orthonormal bases of their respective orthogonal complements.

We define $\alpha := V_*^\top x \in \mathbb{R}^q$ and $\beta := \overline{V}_*^\top x \in \mathbb{R}^{n-q}$ to denote the coordinates of $x \in X$ in the orthonormal bases associated with $V_* \alpha \in \ker(J_*)$ and $\overline{V}_* \beta \in \ker(J_*)^\perp$ respectively. We define coordinate map $\Gamma : \mathbb{R}^q \times \mathbb{R}^{n-q} \rightarrow \mathbb{R}^n$

$$x = \Gamma(\alpha, \beta) := V_* \alpha + \overline{V}_* \beta. \quad (2.5)$$

Finally, we define the map $\Phi^c : \mathbb{R}^q \times \mathbb{R}^{n-q} \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ as the composition of dynamics Φ with coordinate map Γ as:

$$\Phi^c(\alpha, \beta, p) := \Phi(\Gamma(\alpha, \beta), p) = \Phi(V_* \alpha + \overline{V}_* \beta, p).$$

Application of the Implicit Function theorem

The central step of the Lyapunov–Schmidt reduction is the elimination of the β coordinates by applying the Implicit Function theorem to $f(u, v) \equiv W_*^\top \Phi^c(\alpha, \beta, p)$, with $u \equiv (\alpha, p)$ and $v \equiv \beta$. This guarantees the existence of neighbourhoods $\mathcal{N}((\alpha_*, p_*)) \subset \mathbb{R}^q \times \mathbb{R}^m$ and $\mathcal{N}(\beta_*) \subset \mathbb{R}^{n-q}$, together with a unique $\mathcal{C}^2$ map $\varphi : \mathcal{N}((\alpha_*, p_*)) \rightarrow \mathcal{N}(\beta_*)$ that implicitly determines the β coordinates of equilibria near the singular point (x_*, p_*) through $\beta = \varphi(\alpha, p)$. Substituting this map and projecting onto $\text{range}(J_*)^\perp$ yields the reduced system $\psi(\alpha, p) := \overline{W}_*^\top \Phi^c(\alpha, \varphi(\alpha, p), p) = \mathbf{0}_q$, with

$$\Phi^c(\alpha, \varphi(\alpha, p), p) = \mathbf{0}_n \iff \psi(\alpha, p) = \mathbf{0}_q. \quad (2.6)$$

The zeros of ψ define the bifurcation diagram of the reduced system and are locally in one-to-one correspondence with the equilibria of the original n -dimensional system. The construction is summarised in Figure 1.

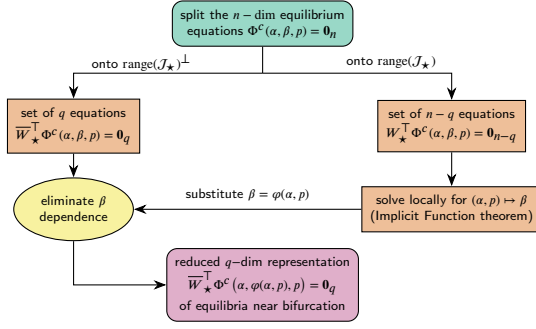


Figure 1: Flowchart depicting the Lyapunov–Schmidt reduction for local bifurcations, as developed in [8, Chapter VII].

3. Second-Order Bounds on the Domain of Validity for Lyapunov–Schmidt reduction

The local graph representation $\beta = \varphi(\alpha, p)$ obtained through the application of the Implicit Function theorem is guaranteed to exist on sufficiently small neighbourhoods of (α_*, p_*) and β_* . In this section, we derive explicit lower bounds on the size of these neighbourhoods and hence on the region over which the reduced and full bifurcation diagrams remain topologically equivalent. By specialising Theorem 2.2 to the Lyapunov–Schmidt reduction formulation introduced in Section 2.3, we obtain computable radii $r_{\parallel}$ and $r_{\perp}$ such that $B((\alpha_*, p_*), r_{\parallel}) \subset \mathcal{N}((\alpha_*, p_*))$ and $B(\beta_*, r_{\perp}) \subset \mathcal{N}(\beta_*)$, providing quantitative guarantees on the domain over which the reduction remains valid. The radius $r_{\parallel}$ quantifies robustness along the critical subspace $\ker(J_*) \times \Lambda$, whereas $r_{\perp}$ quantifies robustness in the transverse directions $\ker(J_*)^{\perp}$.

Theorem 3.1. *Consider the dynamical system described in Section 2.3. Define the following constants:*

$$M_{\parallel} := \|W_*^T D_p \Phi(x_*, p_*)\|, \quad M_{\perp} := \|\Sigma_*^{-1}\| \quad (3.1)$$

where Σ_* is the diagonal matrix of nonzero singular values of $J_* = D_x \Phi(x_*, p_*)$. Define the constant matrices

$$\chi := \begin{bmatrix} V_* & \mathbf{0}_{n \times m} \\ \mathbf{0}_{m \times q} & I_m \end{bmatrix} \in \mathbb{R}^{(n+m) \times (q+m)},$$

$$\bar{\chi} := \begin{bmatrix} \bar{V}_* \\ \mathbf{0}_{m \times (n-q)} \end{bmatrix} \in \mathbb{R}^{(n+m) \times (n-q)},$$

to define the following tensor-valued maps:

$$\begin{aligned} \zeta_{uu}(\alpha, \beta, p) &:= W_*^T D^2 \Phi(x, p) [\chi, \chi] \in \mathbb{R}^{(n-q) \times (q+m) \times (q+m)}, \\ \zeta_{uv}(\alpha, \beta, p) &:= W_*^T D^2 \Phi(x, p) [\chi, \bar{\chi}] \in \mathbb{R}^{(n-q) \times (q+m) \times (n-q)}, \\ \zeta_{vv}(\alpha, \beta, p) &:= W_*^T D^2 \Phi(x, p) [\bar{\chi}, \bar{\chi}] \in \mathbb{R}^{(n-q) \times (n-q) \times (n-q)}, \end{aligned} \quad (3.2)$$

where $x = \Gamma(\alpha, \beta)$ and $D^2 \Phi(x, p) \in \mathbb{R}^{n \times (n+m) \times (n+m)}$ is the Hessian of Φ w.r.t. $(x, p) \in \mathbb{R}^{n+m}$. Given $R_{\parallel}, R_{\perp} > 0$ with $B_{\parallel}(R_{\parallel}) := B((\alpha_*, p_*), R_{\parallel}) \subset \mathbb{R}^q \times \mathbb{R}^m$ and $B_{\perp}(R_{\perp}) := B(\beta_*, R_{\perp}) \subset \mathbb{R}^{n-q}$, define constants K_{uu}, K_{uv}, K_{vv} as the suprema of the respective $\|\zeta\|$ s over these balls jointly:

$$K_{uu} := \sup \{ \|\zeta_{uu}(\alpha, \beta, p)\| : (\alpha, p) \in B_{\parallel}(R_{\parallel}), \beta \in B_{\perp}(R_{\perp}) \},$$

$$\begin{aligned} K_{uv} &:= \sup \{ \|\zeta_{uv}(\alpha, \beta, p)\| : (\alpha, p) \in B_{\parallel}(R_{\parallel}), \beta \in B_{\perp}(R_{\perp}) \}, \\ K_{vv} &:= \sup \{ \|\zeta_{vv}(\alpha, \beta, p)\| : (\alpha, p) \in B_{\parallel}(R_{\parallel}), \beta \in B_{\perp}(R_{\perp}) \}. \end{aligned} \quad (3.3)$$

Then for all $0 < r_{\parallel} < R_{\parallel}$ and $0 < r_{\perp} < R_{\perp}$ satisfying

$$\frac{1}{2} K_{uu} r_{\parallel}^2 + K_{uv} r_{\parallel} r_{\perp} + \frac{1}{2} K_{vv} r_{\perp}^2 < \frac{r_{\perp}}{M_{\perp}} - M_{\parallel} r_{\parallel}, \quad (3.4a)$$

$$K_{uv} r_{\parallel} + K_{vv} r_{\perp} < \frac{1}{M_{\perp}}, \quad (3.4b)$$

simultaneously, there exists a C^2 implicit map $\varphi : B((\alpha_*, p_*), r_{\parallel}) \rightarrow B(\beta_*, r_{\perp})$ that satisfies (2.6).

Proof. The result follows by specialising the quantities defined in Theorem 2.2 to the Lyapunov–Schmidt reduction formulation from Section 2.3. First, the expressions for $M_{\parallel}$ and $M_{\perp}$ are obtained by substituting the Jacobian expressions from Lemma A.1 into the definitions in (2.1), following the proofs in [12, Lemma 3.1, Lemma 3.2]. Then, the ζ 's are obtained by expanding the Hessians $\zeta_{uu} \equiv D_{(\alpha, p)}^2 [W_*^T \Phi^c]$, $\zeta_{uv} \equiv D_{(\alpha, p)} D_{\beta} [W_*^T \Phi^c]$ and $\zeta_{vv} \equiv D_{\beta}^2 [W_*^T \Phi^c]$ in (2.2) as

$$\zeta_{uu} = \begin{bmatrix} W_*^T D_{\alpha}^2 \Phi^c & W_*^T D_{\alpha} D_p \Phi^c \\ W_*^T D_p D_{\alpha} \Phi^c & W_*^T D_p^2 \Phi^c \end{bmatrix}_i \quad (3.5a)$$

$$\zeta_{uv} = \begin{bmatrix} W_*^T D_{\alpha} D_p \Phi^c \\ W_*^T D_p D_{\beta} \Phi^c \end{bmatrix}_i \quad (3.5b)$$

$$\zeta_{vv} = W_*^T D_{\beta}^2 \Phi^c \quad (3.5c)$$

Substituting expressions for the Hessians of $W_*^T \Phi^c$ from Lemmas A.2 and A.3 into the above block forms and grouping terms according to the decompositions induced by χ and $\bar{\chi}$ yields the tensor forms (3.2). Hence, K_{uu}, K_{uv}, K_{vv} in (3.3) coincide with the corresponding Hessian norm suprema in (2.2), and the theorem follows. $\square$

In system-specific applications, exact evaluation of K_{uu}, K_{uv}, K_{vv} may be difficult, as it requires maximising norms of rank-3 tensor-valued maps $\zeta_{uu}, \zeta_{uv}, \zeta_{vv}$ over prescribed neighbourhoods. When these tensors possess exploitable analytical structure, direct optimisation may instead be replaced by simpler upper bounds. The following corollary shows that any computable upper bounds $\tilde{K}_{uu}, \tilde{K}_{uv}, \tilde{K}_{vv}$ on these quantities over $B_{\parallel}(R_{\parallel}) \times B_{\perp}(R_{\perp})$ may be used in place of the exact constants, yielding valid, though potentially more conservative, admissible radii.

Corollary 3.1. *Let $\tilde{K}_{uu} \geq K_{uu}$, $\tilde{K}_{uv} \geq K_{uv}$, and $\tilde{K}_{vv} \geq K_{vv}$. Consider a pair $\rho_{\parallel}, \rho_{\perp} > 0$ with $\rho_{\parallel} < R_{\parallel}$ and $\rho_{\perp} < R_{\perp}$ satisfying (3.4) with the K s replaced by the $\tilde{K}$ s:*

$$\frac{1}{2} \tilde{K}_{uu} \rho_{\parallel}^2 + \tilde{K}_{uv} \rho_{\parallel} \rho_{\perp} + \frac{1}{2} \tilde{K}_{vv} \rho_{\perp}^2 < \frac{\rho_{\perp}}{M_{\perp}} - M_{\parallel} \rho_{\parallel}, \quad (3.6a)$$

$$\tilde{K}_{uv} \rho_{\parallel} + \tilde{K}_{vv} \rho_{\perp} < \frac{1}{M_{\perp}}. \quad (3.6b)$$

Then $\rho_{\parallel}, \rho_{\perp}$ constitute valid bounds for the Lyapunov–Schmidt reduction in the sense of Theorem 3.1.

Proof. Since each $\tilde{K} \geq K$, it follows that for such a pair of $\rho_{\parallel}, \rho_{\perp}$ the left-hand sides of (3.4) evaluated at $(\rho_{\parallel}, \rho_{\perp})$ are bounded above by the corresponding left-hand sides of (3.6). Together with (3.6), this shows that $(\rho_{\parallel}, \rho_{\perp})$ satisfies (3.4). The result therefore follows from Theorem 3.1. $\square$

We now characterise the second-order bounds by studying the geometry of the constraints (3.4) in the $(r_{\parallel}, r_{\perp})$ plane, yielding an explicit description of the admissible region.

3.1. Characterisation of the Second-Order Bounds

As illustrated in Figure 2, (3.4a) defines the interior of a conic section in the first quadrant of the $(r_{\parallel}, r_{\perp})$ plane. The admissible portion is always non-empty, intersecting the $r_{\perp}$ axis for $0 < r_{\perp} < 2/(M_{\perp}K_{vv})$ and the $r_{\parallel}$ axis only at the origin. Similarly, (3.4b) defines the half-plane below the line $K_{uv}r_{\parallel} + K_{vv}r_{\perp} = 1/M_{\perp}$ with positive axis intercepts $r_{\parallel} = 1/(M_{\perp}K_{uv})$ and $r_{\perp} = 1/(M_{\perp}K_{vv})$. The region of admissible $(r_{\parallel}, r_{\perp})$ is naturally the intersection of these sets in the first quadrant.

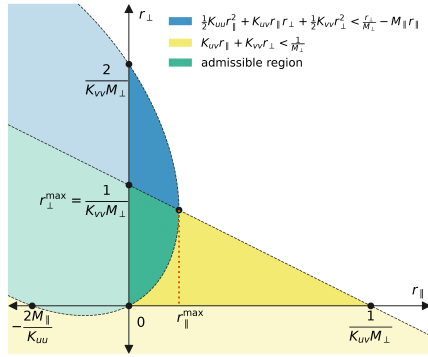


Figure 2: Schematic geometry of the second-order Lyapunov–Schmidt constraints (3.4). Blue and yellow indicate satisfaction of the quadratic (3.4a) and linear (3.4b) inequalities, respectively. Their green overlap represents admissible radius pairs only in the positive quadrant; lighter shading elsewhere illustrates the algebraic continuation. The conic–line intersection determines $r_{\parallel}^{\sup}$, while the line’s vertical intercept gives $r_{\perp}^{\sup}$. The additional restrictions $r_{\parallel} < R_{\parallel}$ and $r_{\perp} < R_{\perp}$ are not shown in this figure.

We now derive explicit expressions for the suprema of radii $r_{\parallel}$ and $r_{\perp}$ admissible under (3.4). We have already established that $r_{\perp}^{\sup} = 1/(K_{vv}M_{\perp})$. To determine $r_{\parallel}^{\sup}$, define

$$F(r_{\parallel}, r_{\perp}) = \frac{1}{2}K_{uu}r_{\parallel}^2 + K_{uv}r_{\parallel}r_{\perp} + \frac{1}{2}K_{vv}r_{\perp}^2 - \frac{r_{\perp}}{M_{\perp}} + M_{\parallel}r_{\parallel},$$

$$G(r_{\parallel}, r_{\perp}) = K_{uv}r_{\parallel} + K_{vv}r_{\perp} - \frac{1}{M_{\perp}},$$

so that the admissible region is given by $F, G < 0$. As depicted in Figure 2, its boundary consists of an arc of $F = 0$ and a segment of $G = 0$. Hence, $r_{\parallel}^{\sup}$ occurs either at a stationary point of $F = 0$ in $r_{\parallel}$, or at an intersection point of $F = 0$ and $G = 0$. Differentiating $F = 0$ implicitly, we get $\left. \frac{dr_{\parallel}}{dr_{\perp}} \right|_{F=0} = -\frac{\partial F/\partial r_{\perp}}{\partial F/\partial r_{\parallel}}$, so such a stationary point satisfies

$$\frac{\partial F}{\partial r_{\perp}} = 0 \implies 0 = K_{uv}r_{\parallel} + K_{vv}r_{\perp} - \frac{1}{M_{\perp}} = G(r_{\parallel}, r_{\perp}).$$

Hence, such a stationary point is an intersection of $F = 0$ and $G = 0$. Solving these simultaneously and rationalising yields the following two roots in $(r_{\parallel}, r_{\perp})$:

$$r_{\parallel} = \frac{1}{M_{\perp} \left(K_{uv} + M_{\parallel}M_{\perp}K_{vv} \pm \sqrt{K_{vv}\Theta} \right)},$$

$$r_{\perp} = \frac{M_{\parallel}M_{\perp}K_{vv} \pm \sqrt{K_{vv}\Theta}}{M_{\perp}K_{vv} \left(K_{uv} + M_{\parallel}M_{\perp}K_{vv} \pm \sqrt{K_{vv}\Theta} \right)},$$

where $\Theta := K_{uu} + 2M_{\parallel}M_{\perp}K_{uv} + M_{\parallel}^2M_{\perp}^2K_{vv}$. Since $\sqrt{K_{vv}\Theta} \geq M_{\parallel}M_{\perp}K_{vv}$, the $(-)$ root cannot lie in the first quadrant. Hence, the $(+)$ root is the relevant solution:

$$r_{\parallel}^{\sup} = \frac{1}{M_{\perp} \left(K_{uv} + M_{\parallel}M_{\perp}K_{vv} + \sqrt{K_{vv}\Theta} \right)}. \quad (3.7)$$

3.2. Significance of $R_{\parallel}$ and $R_{\perp}$

The parameters $R_{\parallel}$ and $R_{\perp}$ define the neighbourhoods over which the Hessian norm bounds K_{uu}, K_{uv}, K_{vv} are evaluated. As these constants are defined as suprema over the prescribed neighbourhoods, they are monotonically non-decreasing with respect to $R_{\parallel}$ and $R_{\perp}$. Consequently, enlarging the search region generally yields more conservative estimates of the admissible radii; cf. (3.4). Conversely, the constraints $r_{\parallel} < R_{\parallel}$ and $r_{\perp} < R_{\perp}$ imply that choosing $R_{\parallel}$ or $R_{\perp}$ too small may itself limit the certified domain of validity. The selection of $R_{\parallel}$ and $R_{\perp}$ therefore represents a trade-off between the neighbourhood over which the analysis is performed and the size of the resulting guaranteed domain.

4. Bifurcations in Hopfield-like Networks

In this section, we examine the bounds presented in the preceding section for a class of networked dynamical systems analogous to continuous-time *Hopfield* neural networks. Such systems arise across a broad range of application domains, including recurrent neural network models [14, 2, 29, 27], models of opinion dynamics and collective decision-making [6, 10, 5, 20], gene regulatory and cellular signalling networks [7, 3], and other interacting systems. Specifically, we consider dynamical systems defined on a graph $\mathcal{G}$ of the canonical form

$$\dot{x} = \Phi(x, p) = -Cx + pAS(x) + b, \quad (4.1)$$

where $x \in \mathbb{R}^n$ is the state, $p \in \mathbb{R}$ is a scalar parameter, $b \in \mathbb{R}^n$ is a constant bias, $A \in \mathbb{R}^{n \times n}$ is the adjacency matrix of $\mathcal{G}$, and $C = \text{diag}(c_1, \dots, c_n) > 0$. The nonlinear function $[S(z)]_i = S_i(z_i)$ applies an elementwise smooth activation function to each input, and its first and second derivatives are given by $[DS(x)]_{ij} = S'_i(x_i)\delta_{ij}$ and $[D^2S(x)]_{ijk} = S''_i(x_i)\delta_{ijk}$.

We consider (4.1) and denote its Jacobian at the singular point by $J_{\star H} := D_x \Phi(x_{\star}, p_{\star}) = -C + p_{\star} A DS(x_{\star})$. We also use the orthonormal basis matrices $V_{\star}, \bar{V}_{\star}, W_{\star}, \bar{W}_{\star}$ induced by $J_{\star H}$, as well as $\Sigma_{\star}$ as described in (2.4). Next, we make the following assumptions:

Assumption 4.1. *There exists $(x_*, p_*) \in \mathbb{R}^n \times \mathbb{R}^{>0}$ such that the following conditions hold:*

1. *The pair (x_*, p_*) is an equilibrium of (4.1), i.e.,*

$$\Phi(x_*, p_*) = -Cx_* + p_* AS(x_*) + b = 0. \quad (4.2)$$

2. *The Jacobian J_{*H} satisfies $\dim \ker J_{*H} = q$, and $\operatorname{Re}(\lambda_j) \neq 0$ for the remaining nonzero eigenvalues.*

We now express the Jacobians and Hessians of (4.1) w.r.t. x and p for subsequent use:

$$D_x \Phi = -C + pA D S(x), \quad D_p \Phi = AS(x), \quad (4.3)$$

$$D_x^2 \Phi = pA D^2 S(x), \quad D_p^2 \Phi = \mathbf{0}_n, \quad (4.4)$$

$$D_x D_p \Phi = A D S(x), \quad D_p D_x \Phi = A D S(x). \quad (4.5)$$

In the following sequence of lemmas, we derive expressions for the quantities defined in Theorem 3.1 required to establish explicit second-order bounds for the validity of the Lyapunov–Schmidt reduction around (x_*, p_*) .

Lemma 4.1.

$$M_{\parallel} = \left\| W_*^T AS(x_*) \right\| = \frac{1}{|p_*|} \left\| W_*^T (Cx_* - b) \right\|, \quad (4.6a)$$

$$M_{\perp} = \left\| \Sigma_*^{-1} \right\|. \quad (4.6b)$$

Proof. Substituting $D_p \Phi(x, p) = AS(x)$ into the definition from (3.1) and using the equilibrium condition (4.2) yields $AS(x_*) = (Cx_* - b)/p_*$, and consequently (4.6a). Identity (4.6b) follows directly from its definition in (3.1). $\square$

Lemma 4.2. $\zeta_{uu} \in \mathbb{R}^{(n-q) \times (q+1) \times (q+1)}$ can be written as

$$\zeta_{uu}(\alpha, \beta, p) = \left[\begin{array}{c|c} pW_*^T A D^2 S(x) [V_*, V_*] & W_*^T A D S(x) V_* \\ \hline W_*^T A D S(x) V_* & \mathbf{0}_{n-q} \end{array} \right]_i$$

Proof. Substituting (4.4) $\rightarrow$ (A.2b), (4.5) $\rightarrow$ (A.3c), (4.5) $\rightarrow$ (A.3d), and (4.4) $\rightarrow$ (A.2a) yields

$$W_*^T D_\alpha^2 \Phi^c = pW_*^T A D^2 S(x) [V_*, V_*] \in \mathbb{R}^{(n-q) \times q \times q}$$

$$W_*^T D_\alpha D_p \Phi^c = W_*^T A D S(x) V_* \in \mathbb{R}^{(n-q) \times q \times 1}$$

$$W_*^T D_p D_\alpha \Phi^c = W_*^T A D S(x) V_* \in \mathbb{R}^{(n-q) \times 1 \times q}$$

$$W_*^T D_p^2 \Phi^c = W_*^T \mathbf{0}_n = \mathbf{0}_{n-q} \in \mathbb{R}^{(n-q) \times 1 \times 1}$$

Collecting these into the blocks of $\zeta_{uu}(\alpha, \beta, p)$ expressed in (3.5a) yields the desired result. $\square$

Lemma 4.3. $\zeta_{uv} \in \mathbb{R}^{(n-q) \times (q+1) \times (n-q)}$ can be written as

$$\zeta_{uv}(\alpha, \beta, p) = \left[\frac{pW_*^T A D^2 S(x) [V_*, \bar{V}_*]}{W_*^T A D S(x) \bar{V}_*} \right]_i$$

Proof. Substituting (4.4) $\rightarrow$ (A.3a) and (4.5) $\rightarrow$ (A.3f),

$$W_*^T D_\alpha D_\beta \Phi^c = pW_*^T A D^2 S(x) [V_*, \bar{V}_*] \in \mathbb{R}^{(n-q) \times q \times (n-q)}$$

$$W_*^T D_p D_\beta \Phi^c = W_*^T A D S(x) \bar{V}_* \in \mathbb{R}^{(n-q) \times 1 \times (n-q)}$$

Collecting these into the blocks of $\zeta_{uv}(\alpha, \beta, p)$ expressed in (3.5b) yields the desired result. $\square$

Lemma 4.4. $\zeta_{vv} \in \mathbb{R}^{(n-q) \times (n-q) \times (n-q)}$ can be written as

$$\zeta_{vv}(\alpha, \beta, p) = pW_*^T A D^2 S(x) [\bar{V}_*, \bar{V}_*]$$

Proof. Substituting (4.4) $\rightarrow$ (A.2c) yields

$$W_*^T D_\beta^2 \Phi^c(\alpha, \beta, p) = W_*^T (pA D^2 S(x)) [\bar{V}_*, \bar{V}_*]$$

which, together with (3.5c) yields the desired expression. $\square$

The expressions derived above hold for general Hopfield networks. We now specialise to the regular graph case, where the derived expressions for ζ_{uu} , ζ_{uv} , ζ_{vv} permit explicit upper bounds on K_{uu} , K_{uv} , K_{vv} , and hence analytical second-order validity bounds under Corollary 3.1.

Evaluation for Regular Graphs

We specialise (4.1) to the unbiased case ($b \equiv \mathbf{0}_n$) over regular graphs by taking $C \equiv dI_n$ for some $d > 0$, $p \equiv u$, $S \equiv \tanh$, and A to be the binary adjacency matrix of a connected, undirected, simple k -regular graph:

$$\dot{x} = \Phi(x, u) = -dx + uA \tanh(x). \quad (4.7)$$

This system undergoes a supercritical pitchfork bifurcation at $(x_*, u_*) = (\mathbf{0}_n, d/k)$ at which branches of equilibria emerge. By (2.5), $(\alpha_*, \beta_*) = (0, \mathbf{0}_{n-1})$.

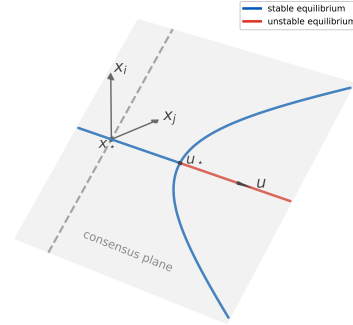


Figure 3: Primary pitchfork bifurcation in Hopfield network (4.7) at $(x_*, u_*) = (\mathbf{0}_n, d/k)$; branches of equilibria emerge from the primary pitchfork within the consensus plane, which is tangent to the consensus (centre) manifold at the bifurcation point.

The Jacobian $J_* := D_x \Phi(x_*, u_*)$ simplifies to

$$J_* = -dI_n + u_* A \operatorname{diag}[\operatorname{sech}^2(x_*)] = \frac{d}{k} (A - kI_n). \quad (4.8)$$

We first characterise the orthonormal basis matrix V_* of $\ker(J_*)$. By the Perron–Frobenius theorem [15, Theorem 8.4.4], A has a simple largest eigenvalue k , with Perron eigenvector $\mathbf{1}_n$. Consequently, $\ker(J_*)$ is one-dimensional ($q = 1$), spanned by $V_* = \mathbf{1}_n / \sqrt{n}$. Since J_* is symmetric, $W_* = \bar{V}_*$ and $V_* = \bar{W}_*$. We also define $\Upsilon_* \in \mathbb{R}^{(n-1) \times (n-1)}$ as the diagonal matrix of all non-Perron eigenvalues ($\neq k$) of A , and let $W_* \in \mathbb{R}^{n \times (n-1)}$ contain the corresponding orthonormal eigenvectors as columns so that $AW_* = W_* \Upsilon_*$. With these simplifications, we now adapt the expressions for the ζ tensors from Lemmas 4.2 to 4.4 to the regular graph case, and characterise their Frobenius norms.

We employ the Frobenius norm rather than the spectral norm, as it permits explicit evaluation of the norms of the rank-3 ζ tensors and facilitates maximisation over the prescribed neighbourhoods. By contrast, the spectral norm does not readily admit analogous analytical or numerical treatment. Even with the Frobenius norm, however, exact maximisation remains analytically intractable for general regular graphs. To obtain explicit bounds, we instead employ Corollary 3.1, using upper bounds on the derivatives of the activation $S = \tanh(\cdot)$ to derive tractable upper bounds on the $\|\zeta\|$ s depending only on u , the network parameters n and k , and the constants $m_1 = 1$ and $m_2 = 4/(3\sqrt{3})$, which bound $S' = \text{sech}^2(\cdot)$ and $S'' = -2 \tanh(\cdot) \text{sech}^2(\cdot)$, respectively. Before establishing bounds on the $\|\zeta\|$ s, we record the following useful lemmas.

Lemma 4.5. $\|W_\star^\top A W_\star\|_F^2 = \|W_\star^\top A\|_F^2 = \|\Upsilon_\star\|_F^2 = k(n-k)$.

Proof. By construction, $W_\star^\top A W_\star = \Upsilon_\star$. Moreover, we have $\|W_\star^\top A\|_F^2 = \|\Upsilon_\star W_\star^\top\|_F^2 = \text{tr}(\Upsilon_\star W_\star^\top W_\star \Upsilon_\star) = \text{tr}(\Upsilon_\star^2) = \|\Upsilon_\star\|_F^2$, since $W_\star^\top W_\star = I_{n-1}$. Finally, $\|\Upsilon_\star\|_F^2 = \sum_{i=2}^n \lambda_i(A)^2 = \|A\|_F^2 - k^2 = nk - k^2 = k(n-k)$, where $\|A\|_F^2 = \sum A_{ij}^2 = nk$ since each node has degree k . $\square$

Lemma 4.6. Let $y \in \mathbb{R}^n$ with $|y_i| \leq m$ for all i . Then,

$$\max_{y \in [-m, m]^n} \|W_\star^\top A \text{diag}(y) W_\star\| = m \|\Upsilon_\star\|,$$

for both the spectral and Frobenius norms.

Proof. Using $W_\star^\top A = \Upsilon_\star W_\star^\top$ together with the submultiplicativity inequality $\|AB\|_F \leq \|A\|_F \|B\|_2$, we have

$$\begin{aligned} \|W_\star^\top A \text{diag}(y) W_\star\|_F &= \|\Upsilon_\star W_\star^\top \text{diag}(y) W_\star\|_F \\ &\leq \|\Upsilon_\star\|_F \|W_\star^\top \text{diag}(y) W_\star\|_2 \\ &\leq \|\Upsilon_\star\|_F \|\text{diag}(y)\|_2 \leq m \|\Upsilon_\star\|_F. \end{aligned}$$

The above also holds for the spectral norm by its submultiplicativity. Since equality is attained at $y = m \mathbf{1}_n$, the bound is tight and coincides with the maximum value of $\|W_\star^\top A \text{diag}(y) W_\star\|$ over the hypercube $[-m, m]^n$. $\square$

Bounding the $\|\zeta\|$ s

We now specialise the expressions for the ζ tensors from Lemmas 4.2 to 4.4 to the connected regular graph setting (4.7), and derive bounds on their Frobenius norms that are independent of α and β and depend only on u .

Starting with ζ_{uu} , we build on Lemma 4.2 to get

$$\zeta_{uu}(\alpha, \beta, u) = \left[\begin{array}{c|c} \frac{u}{n} W_\star^\top A S''(x) & \frac{1}{\sqrt{n}} W_\star^\top A S'(x) \\ \hline \frac{1}{\sqrt{n}} W_\star^\top A S'(x) & \mathbf{0}_{n-1} \end{array} \right]_i.$$

Under the Frobenius norm,

$$\|\zeta_{uu}(\alpha, \beta, u)\|_F^2 = \frac{u^2}{n^2} \|W_\star^\top A S''(x)\|_F^2 + \frac{2}{n} \|W_\star^\top A S'(x)\|_F^2$$

$$\begin{aligned} &\leq \|W_\star^\top A\|_F^2 \left[\frac{u^2}{n^2} \|S''(x)\|_F^2 + \frac{2}{n} \|S'(x)\|_F^2 \right] \\ &\leq [k(n-k)] \left[\frac{u^2}{n^2} (nm_2^2) + \frac{2}{n} (nm_1^2) \right] \\ &= [k(n-k)] \left[\frac{u^2}{n} m_2^2 + 2m_1^2 \right], \end{aligned} \quad (4.9)$$

Next, for ζ_{uv} , building on Lemma 4.3 yields

$$\zeta_{uv}(\alpha, \beta, u) = \left[\frac{\frac{u}{\sqrt{n}} W_\star^\top A \text{diag}(S''(x)) \bar{V}_\star}{W_\star^\top A \text{diag}(S'(x)) \bar{V}_\star} \right]_i \quad (4.10)$$

Under the Frobenius norm,

$$\|\zeta_{uv}(\alpha, \beta, u)\|_F^2 = \frac{u^2}{n} \|W_\star^\top A \text{diag}(S''(x)) \bar{V}_\star\|_F^2 + \|W_\star^\top A \text{diag}(S'(x)) \bar{V}_\star\|_F^2.$$

Applying Lemma 4.6 to the above expression yields

$$\begin{aligned} \|\zeta_{uv}(\alpha, \beta, u)\|_F^2 &\leq \|\Upsilon_\star\|_F^2 \left[\frac{u^2}{n} m_2^2 + m_1^2 \right] \\ &= [k(n-k)] \left[\frac{u^2}{n} m_2^2 + m_1^2 \right], \end{aligned} \quad (4.11)$$

Finally, for ζ_{vv} , using Lemma 4.4 in index notation,

$$\begin{aligned} [\zeta_{vv}(\alpha, \beta, u)]_{ijk} &= u W_{\star \mu i} A_{\mu \nu} S''_v(x_v) \bar{V}_{\star \nu j} \bar{V}_{\star \nu k} \\ &= u \Upsilon_{\star ii} S''_v(x_v) W_{\star \nu i} W_{\star \nu j} W_{\star \nu k}, \end{aligned}$$

for $i, j, k \in \{1, \dots, n-1\}$ and $\mu, \nu \in \{1, \dots, n\}$, since $\bar{V}_\star = W_\star$ and $W_{\star \mu i} A_{\mu \nu} = (W_\star^\top A)_{i\nu} = (\Upsilon_\star W_\star^\top)_{i\nu} = \Upsilon_{\star ii} W_{\star \nu i}$. Expanding the Frobenius norm gives

$$\|\zeta_{vv}(\alpha, \beta, u)\|_F^2 = u^2 \sum_{i,j,k} \Upsilon_{\star ii}^2 \left(\sum_{\nu=1}^n S''_v(x_v) W_{\star \nu i} W_{\star \nu j} W_{\star \nu k} \right)^2.$$

In Appendix B, we show that the summation

$$\sum_{j,k} \left(\sum_{\nu=1}^n S''_v(x_v) W_{\star \nu i} W_{\star \nu j} W_{\star \nu k} \right)^2 \leq m_2^2 (1 - 1/n).$$

Substituting this into the above yields

$$\begin{aligned} \|\zeta_{vv}(\alpha, \beta, u)\|_F^2 &\leq (um_2)^2 \left(1 - \frac{1}{n} \right) \sum_{i=1}^{n-1} \Upsilon_{\star ii}^2 \\ &= (um_2)^2 k(n-k) (1 - 1/n), \end{aligned} \quad (4.12)$$

since $\sum_i \Upsilon_{\star ii}^2 = \|\Upsilon_\star\|_F^2 = k(n-k)$.

We are now ready to state explicit second-order bounds for the primary bifurcation in (4.7) at $(x_\star, u_\star) = (\mathbf{0}_n, d/k)$.

Theorem 4.1. Consider the unbiased Hopfield-like dynamics in (4.7). For fixed $R_\parallel > 0$, define the constants

$$\begin{aligned} \tilde{K}_{uu} &= \sqrt{k(n-k)} \sqrt{\frac{(d/k + R_\parallel)^2}{n} m_2^2 + 2m_1^2}, \\ \tilde{K}_{uv} &= \sqrt{k(n-k)} \sqrt{\frac{(d/k + R_\parallel)^2}{n} m_2^2 + m_1^2}, \\ \tilde{K}_{vv} &= m_2(d/k + R_\parallel) \sqrt{k(n-k)(1 - 1/n)}. \end{aligned}$$

Then for all $0 < \rho_{\parallel} < R_{\parallel}$ and $0 < \rho_{\perp}$ satisfying

$$\frac{1}{2}\tilde{K}_{uu}\rho_{\parallel}^2 + \tilde{K}_{uv}\rho_{\parallel}\rho_{\perp} + \frac{1}{2}\tilde{K}_{vv}\rho_{\perp}^2 < \frac{\rho_{\perp}}{\frac{k}{d}\sqrt{\sum_{i=2}^n \frac{1}{(k-\lambda_i)^2}}},$$

$$\tilde{K}_{uv}\rho_{\parallel} + \tilde{K}_{vv}\rho_{\perp} < \frac{1}{\frac{k}{d}\sqrt{\sum_{i=2}^n \frac{1}{(k-\lambda_i)^2}}},$$

where $k = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_n$ are the ordered eigenvalues of A , there exists a smooth implicit map $\varphi : B((0, d/k), \rho_{\parallel}) \rightarrow B(\mathbf{0}_{n-1}, \rho_{\perp})$ that satisfies (2.6) for the dynamics in (4.7).

Proof. The result follows by verifying the constants appearing in Theorem 3.1. First, $M_{\parallel} = \|W_{\star}^T A \tanh(\mathbf{0}_n)\|_F = 0$. Next, we evaluate $M_{\perp}$ by computing $\Sigma_{\star}$ using (4.8) as

$$\Sigma_{\star} = W_{\star}^T J_{\star} \bar{V}_{\star} = \frac{d}{k} W_{\star}^T (A - kI_n) W_{\star} = \frac{d}{k} (\Upsilon_{\star} - kI_{n-1})$$

$$\Rightarrow M_{\perp} = \|\Sigma_{\star}^{-1}\|_F = \frac{k}{d} \sqrt{\sum_{i=2}^n \frac{1}{(k-\lambda_i)^2}}$$

Finally, applying Corollary 3.1 to the upper bounds on $\|\zeta\|$ s from (4.9), (4.11), and (4.12), together with $|u| < d/k + R_{\parallel}$ yields the required expressions for $\tilde{K}_{uu}$, $\tilde{K}_{uv}$, $\tilde{K}_{vv}$, and the claimed bounds follow immediately. $\square$

5. Comparison with First-Order Bounds

Having derived the second-order bounds, we now compare them with the first-order bounds of [12], recalled below in Theorems 5.1 and 5.2. We begin in Section 5.1 with a comparison of the maximal admissible radii for Hopfield dynamics on complete graphs, providing preliminary intuition for the relative conservativeness of the second-order bounds. In Section 5.2, a general degeneracy condition extends the inclusion result to all connected regular graphs. Finally, we show via a simple counterexample that this inclusion cannot be asserted universally for arbitrary dynamical systems.

Theorem 5.1. [12, Theorem 3.3] Building upon the definitions from Section 2.3, we define the constants

$$M_{\parallel} = \|W_{\star}^T D_p \Phi(x_{\star}, p_{\star})\|, \quad M_{\perp} = \|\Sigma_{\star}^{-1}\|, \quad (5.1)$$

and maps $\xi_{\parallel} : \mathbb{R}^q \times \mathbb{R}^m \rightarrow \mathbb{R}^{(n-q) \times (q+m)}$ and $\xi_{\perp} : \mathbb{R}^q \times \mathbb{R}^{n-q} \times \mathbb{R}^m \rightarrow \mathbb{R}^{(n-q) \times (n-q)}$ as

$$\xi_{\parallel}(\alpha, p) := W_{\star}^T [\xi_{\parallel\alpha}(\alpha, p) \quad \xi_{\parallel p}(\alpha, p)], \quad (5.2a)$$

$$\xi_{\perp}(\alpha, \beta, p) := W_{\star}^T D_x \Phi(\Gamma(\alpha, \beta), p) \bar{V}_{\star} - \Sigma_{\star}, \quad (5.2b)$$

where $\xi_{\parallel\alpha}(\alpha, p) = D_x \Phi(\Gamma(\alpha, \beta_{\star}), p) V_{\star}$ and $\xi_{\parallel p}(\alpha, p) = D_p \Phi(\Gamma(\alpha, \beta_{\star}), p) - D_p \Phi(x_{\star}, p_{\star})$. We introduce shorthand notations $B_{\parallel}(r_{\parallel}) := B((\alpha_{\star}, p_{\star}), r_{\parallel}) \subset \mathbb{R}^q \times \mathbb{R}^m$ and $B_{\perp}(r_{\perp}) := B(\beta_{\star}, r_{\perp}) \subset \mathbb{R}^{n-q}$ to denote open balls centred at the singular point. Finally, for $r_{\parallel}, r_{\perp} > 0$, define

$$L_{\parallel}(r_{\parallel}) := \sup_{(\alpha, p) \in B_{\parallel}(r_{\parallel})} \|\xi_{\parallel}(\alpha, p)\|, \quad (5.3a)$$

$$L_{\perp}(r_{\parallel}, r_{\perp}) := \sup_{\substack{(\alpha, p) \in B_{\parallel}(r_{\parallel}) \\ \beta \in B_{\perp}(r_{\perp})}} \|\xi_{\perp}(\alpha, \beta, p)\|. \quad (5.3b)$$

Then, for all $r_{\parallel}, r_{\perp} > 0$ satisfying

$$L_{\parallel}(r_{\parallel})r_{\parallel} + L_{\perp}(r_{\parallel}, r_{\perp})r_{\perp} \leq \frac{r_{\perp}}{M_{\perp}} - M_{\parallel}r_{\parallel} \quad (5.4)$$

there exists a smooth implicit map

$\varphi : B((\alpha_{\star}, p_{\star}), r_{\parallel}) \rightarrow B(\beta_{\star}, r_{\perp})$ that satisfies (2.6).

Theorem 5.2. Consider (4.7) over a connected regular graph with n nodes of degree k . Sufficient first-order bounds for the Lyapunov–Schmidt reduction from Theorem 5.1 around the pitchfork bifurcation at $(x_{\star}, u_{\star}) = (\mathbf{0}_n, d/k)$ are given by

$$\|\Upsilon_{\star}\| \|\Sigma_{\star}^{-1}\| \left(r_{\parallel} + \max \left\{ \frac{d}{k} - r_{\parallel}, 0 \right\} \tanh^2 \left(\frac{r_{\parallel}}{\sqrt{n}} + \sqrt{\frac{n-1}{n}} r_{\perp} \right) \right) < 1,$$

where $r_{\parallel} > 0$ and $r_{\perp} > 0$ are the radii along $\ker(J_{\star}) \times \mathbb{R}$ and $\ker(J_{\star})^{\perp}$, respectively. Whenever the graph satisfies $\frac{d}{k} \|\Sigma_{\star}^{-1}\| \|\Upsilon_{\star}\| \leq 1$, these bounds simplify to

$$0 < r_{\parallel} < \frac{1}{\|\Upsilon_{\star}\| \|\Sigma_{\star}^{-1}\|} \quad \text{and} \quad r_{\perp} > 0. \quad (5.5)$$

Proof. From [12, Section V], we have $M_{\parallel} = L_{\parallel}(r_{\parallel}) = 0$ for both the spectral and Frobenius norms, and

$$\|\xi_{\perp}(\alpha, \beta, u)\| = \|W_{\star}^T A \operatorname{diag} \left(u \operatorname{sech}^2(\Gamma(\alpha, \beta)) - \frac{d}{k} \right) W_{\star}\|.$$

Using $|u - d/k| < r_{\parallel}$ (arising from the domain's radius constraint) and $\operatorname{sech}^2(\cdot) = 1 - \tanh^2(\cdot)$, we first obtain

$$\begin{aligned} \left| u \operatorname{sech}^2(\Gamma_i) - \frac{d}{k} \right| &\leq \left| u - \frac{d}{k} \right| \operatorname{sech}^2(\Gamma_i) + \frac{d}{k} \tanh^2(\Gamma_i) \\ &\leq r_{\parallel} + \left(\frac{d}{k} - r_{\parallel} \right) \tanh^2(\Gamma_i) \\ &\leq r_{\parallel} + \max \left\{ \frac{d}{k} - r_{\parallel}, 0 \right\} \tanh^2(\Gamma_i). \end{aligned}$$

Applying Cauchy–Schwarz elementwise to Γ_i from (2.5),

$$|\Gamma_i(\alpha, \beta)| \leq \frac{|\alpha|}{\sqrt{n}} + \sqrt{\frac{n-1}{n}} \|\beta\|_2 \leq \frac{r_{\parallel}}{\sqrt{n}} + \sqrt{\frac{n-1}{n}} r_{\perp}.$$

Combining these with Lemma 4.6 and substituting the above into $M_{\perp} L_{\perp}(r_{\parallel}, r_{\perp}) < 1$ yields the first stated condition. Finally, if $\frac{d}{k} \|\Upsilon_{\star}\| \|\Sigma_{\star}^{-1}\| \leq 1$, this condition holds automatically for $0 < r_{\parallel} < d/k$, while for $r_{\parallel} \geq d/k$ it reduces to $r_{\parallel} \|\Upsilon_{\star}\| \|\Sigma_{\star}^{-1}\| < 1$, which completes the proof. $\square$

Remark 5.1 (Choice of norm). The first-order bounds in Theorem 5.2 have the same analytical form for the spectral and Frobenius norms, but their values and the satisfaction of $\frac{d}{k} \|\Sigma_{\star}^{-1}\| \|\Upsilon_{\star}\| \leq 1$ depend on the choice of norm and the graph spectrum. An alternative proof based on monotonicity arguments can be used to establish the bound (5.5) for all connected regular graphs; we choose to present the more conservative condition here to emphasise the relationship between the first-order bounds of Theorem 5.2 and the second-order bounds presented in Theorem 4.1. Our numerical experiments use the Frobenius norm.

5.1. Preliminary Comparison for Complete Graphs

Before addressing the general case, it is instructive to consider a setting in which the first-order bound takes the simple, decoupled form of (5.5). By Theorem 5.2, this simplification is guaranteed whenever $\frac{d}{k} \|\Sigma_\star^{-1}\| \|\Upsilon_\star\| \leq 1$, a condition satisfied by every complete graph under both the Frobenius and spectral norms. Restricting attention to this case for now lets us compare the first and second-order maximal radii directly in closed form, giving preliminary insight into the relative conservativeness of the two bounds.

Since the first-order validity region is unbounded in $r_\perp$, whereas the second-order validity region satisfies $r_\perp^{\text{sup}} = 1/(M_\perp K_{vv})$, it remains only to compare the admissible ranges of $r_\parallel$. Specifically, it suffices to show that the maximal admissible $r_\parallel$ under the second-order bounds does not exceed $1/(\|\Upsilon_\star\| \|\Sigma_\star^{-1}\|)$. Substituting $M_\parallel = 0$ into (3.7) yields

$$r_\parallel^{\text{sup}} = \frac{1}{M_\perp(K_{uv} + \sqrt{K_{uu}K_{vv}})}.$$

Since $M_\perp = \|\Sigma_\star^{-1}\|$, it therefore suffices to establish

$$\begin{aligned} \frac{1}{M_\perp(K_{uv} + \sqrt{K_{uu}K_{vv}})} &\leq \frac{1}{M_\perp \|\Upsilon_\star\|} \\ \iff K_{uv} + \sqrt{K_{uu}K_{vv}} &\geq \|\Upsilon_\star\| \end{aligned} \quad (5.6)$$

By the definition of K_{uv} in (3.3), $K_{uv} \geq \|\zeta_{uv}(0, \mathbf{0}_{n-1}, u_\star)\|$. Evaluating $\zeta_{uv}(0, \mathbf{0}_{n-1}, u_\star)$ from (4.10), we obtain

$$\zeta_{uv}(0, \mathbf{0}_{n-1}, u_\star) = \left[\frac{\mathbf{0}_{(n-1) \times (n-1)}}{W_\star^\top A W_\star} \right]_i$$

since $S''(\mathbf{0}_n) = \mathbf{0}_n$ and $S'(\mathbf{0}_n) = \mathbf{1}_n$ for $S \equiv \tanh$. Hence,

$$\|\zeta_{uv}(0, \mathbf{0}_{n-1}, u_\star)\| = \|W_\star^\top A W_\star\| = \|\Upsilon_\star\|$$

for both the spectral and Frobenius norms. Therefore, $K_{uv} \geq \|\Upsilon_\star\|$. Since $K_{uu}, K_{vv} \geq 0$, we have $K_{uv} + \sqrt{K_{uu}K_{vv}} \geq K_{uv} \geq \|\Upsilon_\star\|$, which establishes (5.6). Consequently, for complete graphs, the second-order admissible region is contained in the first-order strip (5.5). We show next that this conclusion extends to every connected regular graph.

5.2. A General Comment on the Conservativeness

The comparison of Section 5.1 was carried out only for complete graphs, where the first-order bound takes the simple, decoupled form of Equation (5.5). To extend the inclusion result to the remaining connected regular graphs, we instead compare the general first and second-order conditions (5.4) and (3.4) directly. This comparison identifies a degeneracy condition that holds for Hopfield dynamics on every connected regular graph and also applies beyond this network class. Finally, we show via a simple counterexample that outside this degenerate regime, no such inclusion can be expected to hold universally for arbitrary systems.

The quantities $L_\parallel(r_\parallel), L_\perp(r_\parallel, r_\perp)$ defined in the first-order bounds in (5.3) and K_{uu}, K_{uv}, K_{vv} of the second-order bounds in (3.3) are not independent: since $\xi_\parallel(\alpha, p), \xi_\perp(\alpha, \beta, p)$ are themselves increments of a Jacobian block of Φ from its value at $(\alpha_\star, \beta_\star, p_\star)$, the mean value theorem bounds them directly by the corresponding Hessian suprema, in the spirit of the argument relating the two Implicit Function theorem formulations in [17].

Theorem 5.3. *For all $0 < r_\parallel < R_\parallel$ and $0 < r_\perp < R_\perp$,*

$$L_\parallel(r_\parallel) \leq K_{uu}r_\parallel, \quad L_\perp(r_\parallel, r_\perp) \leq K_{uv}r_\parallel + K_{vv}r_\perp. \quad (5.7)$$

Proof. Write $\eta(\alpha, p) := [D_\alpha \phi(\alpha, \beta_\star, p) \ D_p \phi(\alpha, \beta_\star, p)]$, where $\phi := W_\star^\top \Phi^c$. Since $D_\alpha \phi(\alpha_\star, \beta_\star, p_\star) = W_\star^\top J_\star V_\star = \mathbf{0}$, comparing with (5.2a) gives $\xi_\parallel(\alpha, p) = \eta(\alpha, p) - \eta(\alpha_\star, p_\star)$. Integrating along

$$(\alpha(t), p(t)) := (\alpha_\star, p_\star) + t[(\alpha, p) - (\alpha_\star, p_\star)] \quad t \in [0, 1],$$

the fundamental theorem of calculus and (3.5a) give

$$\xi_\parallel(\alpha, p) = \int_0^1 \zeta_{uu}(\alpha(t), \beta_\star, p(t)) [(\alpha, p) - (\alpha_\star, p_\star)] dt,$$

so $\|\xi_\parallel(\alpha, p)\| \leq K_{uu}\|(\alpha, p) - (\alpha_\star, p_\star)\| \leq K_{uu}r_\parallel$ for $(\alpha, p) \in B_\parallel(r_\parallel)$; taking the supremum gives the first bound. A similar construction using the fundamental theorem of calculus and (3.5b) shows that $L_\perp(r_\parallel, r_\perp) \leq K_{uv}r_\parallel + K_{vv}r_\perp$. $\square$

In the following corollary, we use Theorem 5.3 to establish a general sufficient condition for inclusion based on the degeneracy $M_\parallel = L_\parallel(r_\parallel) = 0$. This generalises the preliminary complete-graph comparison in Section 5.1 to all connected regular graphs, and further applies to other systems exhibiting the same degeneracy.

Corollary 5.1. *If $M_\parallel = L_\parallel(r_\parallel) = 0$, then every $(r_\parallel, r_\perp)$ satisfying (3.4) also satisfies (5.4); the second-order domain of validity is contained in the first-order domain.*

Proof. With $M_\parallel = 0$ and $L_\parallel(r_\parallel) = 0$, (5.4) reduces to $L_\perp(r_\parallel, r_\perp) < 1/M_\perp$. Combining (3.4b) with Theorem 5.3, $L_\perp(r_\parallel, r_\perp) \leq K_{uv}r_\parallel + K_{vv}r_\perp < 1/M_\perp$, which is precisely this condition. $\square$

For Hopfield dynamics (4.7) on any connected regular graph, $M_\parallel = L_\parallel(r_\parallel) = 0$, as established in the proof of Theorem 5.2. Hence, Corollary 5.1 shows that every radius pair certified by Theorem 3.1 is also certified by Theorem 5.1, irrespective of whether the graph satisfies $\frac{d}{k} \|\Sigma_\star^{-1}\| \|\Upsilon_\star\| \leq 1$.

Away from this degenerate regime, however, *neither* bound dominates the other in general; which one is tighter depends on Φ itself. Consider $f(u, v) = v + u^p$ at $(u_0, v_0) = (0, 0)$ with $p \in \{2, 3, \dots\}$. Fix arbitrarily large $R_u, R_v > 0$, and compare the two certificates for $0 < r_u < R_u$ and $0 < r_v < R_v$. Here, $M_u = \|D_u f(0, 0)\| = 0$, $M_v = 1$, and $D_v f \equiv 1$, so $L_v \equiv 0$; only the u -direction contributes to the radius conditions, with $L_u(r_u) = \sup_{|u| < r_u} |pu^{p-1}| = pr_u^{p-1}$. The

first-order condition in [17, Corollary 3.8], which specialises to (5.4) in the Lyapunov–Schmidt reduction setting, yields

$$r_v > pr_u^p. \quad (5.8)$$

For the second-order bounds of Theorem 2.2, $K_{uv} = K_{vv} = 0$ and $K_{uu} = p(p-1)R_u^{p-2}$. Since (2.3b) holds automatically, (2.3a) yields

$$r_v > \frac{1}{2}p(p-1)R_u^{p-2}r_u^2. \quad (5.9)$$

The ratio of the second-order threshold to the first-order threshold is $\frac{p-1}{2}(R_u/r_u)^{p-2}$. Since $0 < r_u < R_u$, this ratio equals $1/2$ for $p = 2$ and exceeds 1 for every $p \geq 3$. Thus, the second-order region is strictly larger for $p = 2$, whereas the first-order region is strictly larger for every $p \geq 3$. Corollary 5.1 therefore cannot be strengthened to an unconditional statement: the two families of bounds are complementary certificates to be compared case by case, rather than successive refinements of one another.

5.3. Qualitative Differences between the Bounds

Beyond the relative tightness examined above, the two bound families differ qualitatively in their structures.

First, in computational structure: the first-order quantities $L_{\parallel}(r_{\parallel}), L_{\perp}(r_{\parallel}, r_{\perp})$ are suprema taken over the neighbourhoods determined by $(r_{\parallel}, r_{\perp})$ being certified in (5.3), so verifying (5.4) at a candidate radius pair requires optimising over that same pair, and obtaining the maximal admissible $(r_{\parallel}, r_{\perp})$ in general amounts to a running search. The second-order quantities K_{uu}, K_{uv}, K_{vv} , on the other hand, are evaluated once over an a priori fixed region $(R_{\parallel}, R_{\perp})$, after which (3.4) is a static algebraic system with a closed-form solution, as in Section 3.1. Corollary 3.1 extends this advantage further, allowing any computable upper bound on K_{uu}, K_{uv}, K_{vv} to be used in place of the exact suprema without loss of validity.

Second, in the geometry of the certified region: the quadratic structure of (3.4) guarantees a bounded admissible set with explicit finite radii, i.e., $r_{\parallel}^{\sup}$ and $r_{\perp}^{\sup}$, whereas the first-order region can be unbounded in a direction, as it is for complete graphs in Theorem 5.2, where $r_{\perp}$ is unconstrained.

These distinctions, together with the fact that neither bound dominates the other outside the degenerate regime of Corollary 5.1, indicate that the choice between the two is best made on computational, rather than purely analytical, grounds: the first-order bounds are preferable when $L_{\parallel}(r_{\parallel}), L_{\perp}(r_{\parallel}, r_{\perp})$ are cheap to evaluate or optimise, and the second-order bounds are preferable when Hessian estimates are available, or when an explicitly described, easily visualised closed-form region, as in Figure 2, is preferred over a tighter but implicit one.

6. Numerical Results and Visualisations

We now turn to a numerical investigation of the second-order bounds for the primary bifurcation in the Hopfield-like dynamics (4.7) on regular graphs with $d = 1$. We first outline the computation of the certified radii for both the *exact* and *approximate* cases in the sense of Corollary 3.1.

For the *exact* bounds, the quantities K_{uu}, K_{uv}, K_{vv} defined in (3.3) depend on the search radii $R_{\parallel}$ and $R_{\perp}$, inducing $r_{\parallel}^{\sup}(R_{\parallel}, R_{\perp})$ and $r_{\perp}^{\sup}(R_{\parallel}, R_{\perp})$ as evaluated in Section 5.1. As discussed in Section 3.2, the search radii themselves clip the certified domain whenever they are too small. Hence, for a fixed graph, we define the reported radii $r_{\parallel}^*$ and $r_{\perp}^*$ as the suprema of the solution sets of the fixed-point equations $r_{\parallel}^{\sup}(R_{\parallel}, R_{\perp}) = R_{\parallel}$ and $r_{\perp}^{\sup}(R_{\parallel}, R_{\perp}) = R_{\perp}$, respectively. Similarly, for the *approximate* bounds, the reported radii $\rho_{\parallel}^*$ and $\rho_{\perp}^*$ are the largest solutions of their respective fixed-point equations, obtained using the analytical upper bounds $\tilde{K}_{uu}, \tilde{K}_{uv}, \tilde{K}_{vv}$, which depend only on $R_{\parallel}$.

For each (n, k) pair, the reported radii are averaged over up to 10 distinct k -regular graphs on n vertices; if fewer than 10 non-isomorphic graphs exist, all such graphs are included.

Figure 4 illustrates how the exact second-order bounds $r_{\parallel}^*, r_{\perp}^*$ vary with graph size n and degree k . Figure 5 depicts the conservativeness introduced by Corollary 3.1 by visualising the ratio r^*/ρ^* between the exact and approximate bounds.

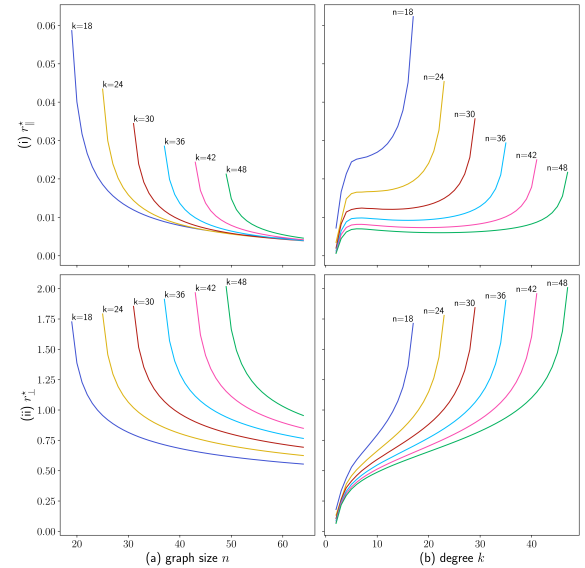


Figure 4: Trends in (i) $r_{\parallel}^*$ and (ii) $r_{\perp}^*$ with graph size and degree: (a) fixed degree k ; (b) fixed graph size n .

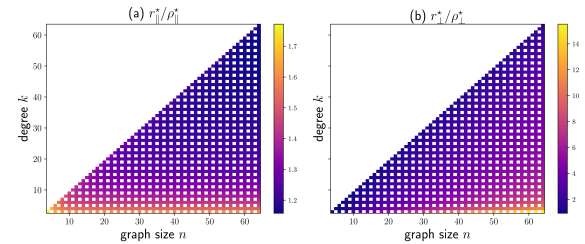


Figure 5: Comparison of the exact and approximate bounds via heatmaps of their ratios (a) $r_{\parallel}^*/\rho_{\parallel}^*$ and (b) $r_{\perp}^*/\rho_{\perp}^*$ across graph sizes $n \leq 64$ and admissible degrees $3 \leq k \leq n-1$; the approximate bounds are most conservative for sparse graphs (small k) but track the exact bounds closely otherwise.

A. Partial Derivatives of ϕ

This section collects the derivations of expressions for the Jacobians and Hessians of $\phi(\alpha, \beta, p) := W_\star^\top \Phi^c(\alpha, \beta, p)$ w.r.t. α , β , and p in terms of those of $\Phi(x, p)$ w.r.t. x and p and the orthonormal basis matrices $V_\star, \bar{W}_\star \in \mathbb{R}^{n \times q}$ and $\bar{V}_\star, W_\star \in \mathbb{R}^{n \times (n-q)}$ defined in (2.4). We use the Einstein summation convention for handling tensors wherever necessary, wherein repeated *dummy* indices are summed over their entire range (both dummy and free indices are subscripted).

Lemma A.1. *The Jacobians of ϕ can be expressed as*

$$D_p \phi = W_\star^\top D_p \Phi(x, p) \in \mathbb{R}^{(n-q) \times m}, \quad (\text{A.1a})$$

$$D_\alpha \phi = W_\star^\top D_x \Phi(x, p) V_\star \in \mathbb{R}^{(n-q) \times q}, \quad (\text{A.1b})$$

$$D_\beta \phi = W_\star^\top D_x \Phi(x, p) \bar{V}_\star \in \mathbb{R}^{(n-q) \times (n-q)}, \quad (\text{A.1c})$$

where $x = \Gamma(\alpha, \beta) = V_\star \alpha + \bar{V}_\star \beta$ as defined in (2.5).

Proof. (A.1b) and (A.1c) follow directly from the chain rule,

$$\begin{aligned} D_\alpha \phi(\alpha, \beta, p) &= W_\star^\top D_x \Phi(x, p) D_\alpha x = W_\star^\top D_x \Phi(x, p) V_\star, \\ D_\beta \phi(\alpha, \beta, p) &= W_\star^\top D_x \Phi(x, p) D_\beta x = W_\star^\top D_x \Phi(x, p) \bar{V}_\star, \end{aligned}$$

while (A.1a) follows directly. $\square$

Lemma A.2. *The strict partials of ϕ can be expressed as*

$$D_p^2 \phi = W_\star^\top D_p^2 \Phi(x, p) \in \mathbb{R}^{(n-q) \times m \times m}, \quad (\text{A.2a})$$

$$D_\alpha^2 \phi = W_\star^\top D_x^2 \Phi(x, p) [V_\star, V_\star] \in \mathbb{R}^{(n-q) \times q \times q}, \quad (\text{A.2b})$$

$$D_\beta^2 \phi = W_\star^\top D_x^2 \Phi(x, p) [\bar{V}_\star, \bar{V}_\star] \in \mathbb{R}^{(n-q) \times (n-q) \times (n-q)}. \quad (\text{A.2c})$$

Proof. (A.2a) is obtained by differentiating (A.1a) w.r.t. p again. (A.2b) follows from differentiating $\phi_i(\alpha, \beta, p) = W_{\star ai} \Phi_a(x, p)$ twice with respect to α to obtain

$$[D_\alpha^2 \phi(\alpha, \beta, p)]_{ijk} = W_{\star ai} \frac{\partial^2 \Phi_a}{\partial x_b \partial x_c}(x, p) \frac{\partial x_b}{\partial \alpha_j} \frac{\partial x_c}{\partial \alpha_k},$$

and using $x_h = V_{\star hj} \alpha_j + \bar{V}_{\star hj} \beta_j \implies \frac{\partial x_h}{\partial \alpha_j} = V_{\star hj}$ to get

$$[D_\alpha^2 \phi(\alpha, \beta, p)]_{ijk} = W_{\star ai} [D_x^2 \Phi(x, p)]_{abc} V_{\star bj} V_{\star ck},$$

which is just $W_\star^\top D_x^2 \Phi(x, p) [V_\star, V_\star]$ in multilinear notation. (A.2c) follows analogously by using $\frac{\partial x_h}{\partial \beta_j} = \bar{V}_{\star hj}$. $\square$

Lemma A.3. *The mixed partials of ϕ can be expressed as*

$$D_\alpha D_\beta \phi = W_\star^\top D_x^2 \Phi(x, p) [V_\star, \bar{V}_\star] \in \mathbb{R}^{(n-q) \times q \times (n-q)}, \quad (\text{A.3a})$$

$$D_\beta D_\alpha \phi = W_\star^\top D_x^2 \Phi(x, p) [\bar{V}_\star, V_\star] \in \mathbb{R}^{(n-q) \times (n-q) \times q}, \quad (\text{A.3b})$$

$$D_\alpha D_p \phi = W_\star^\top D_x D_p \Phi(x, p) [V_\star, I_m] \in \mathbb{R}^{(n-q) \times q \times m}, \quad (\text{A.3c})$$

$$D_p D_\alpha \phi = W_\star^\top D_p D_x \Phi(x, p) [I_m, V_\star] \in \mathbb{R}^{(n-q) \times m \times q}, \quad (\text{A.3d})$$

$$D_\beta D_p \phi = W_\star^\top D_x D_p \Phi(x, p) [\bar{V}_\star, I_m] \in \mathbb{R}^{(n-q) \times (n-q) \times m}, \quad (\text{A.3e})$$

$$D_p D_\beta \phi = W_\star^\top D_p D_x \Phi(x, p) [I_m, \bar{V}_\star] \in \mathbb{R}^{(n-q) \times m \times (n-q)}. \quad (\text{A.3f})$$

Proof. (A.3a) is obtained by differentiating $\phi_i(\alpha, \beta, p) = W_{\star ai} \Phi_a(x, p)$ first with respect to β_k and then α_j :

$$[D_\alpha D_\beta \phi]_{ijk} = W_{\star ai} \frac{\partial^2 \Phi_a}{\partial x_b \partial x_c}(x, p) \frac{\partial x_b}{\partial \alpha_j} \frac{\partial x_c}{\partial \beta_k}.$$

Substituting $\frac{\partial x_b}{\partial \alpha_j} = V_{\star bj}$ and $\frac{\partial x_c}{\partial \beta_k} = \bar{V}_{\star ck}$ above yields

$$\begin{aligned} [D_\alpha D_\beta \phi]_{ijk} &= W_{\star ai} [D_x^2 \Phi(x, p)]_{abc} V_{\star bj} \bar{V}_{\star ck} \\ \implies D_\alpha D_\beta \phi &= W_\star^\top D_x^2 \Phi(x, p) [V_\star, \bar{V}_\star], \end{aligned}$$

in multilinear notation. Reversing the differentiation order,

$$\begin{aligned} [D_\beta D_\alpha \phi]_{ijk} &= W_{\star ai} [D_x^2 \Phi(x, p)]_{abc} \bar{V}_{\star bj} V_{\star ck} \\ \implies D_\beta D_\alpha \phi &= W_\star^\top D_x^2 \Phi(x, p) [\bar{V}_\star, V_\star], \end{aligned}$$

yielding (A.3b). To show (A.3c), we differentiate $\phi_i(\alpha, \beta, p)$ first with respect to p_k and then α_j :

$$\begin{aligned} [D_\alpha D_p \phi]_{ijk} &= W_{\star ai} \frac{\partial^2 \Phi_a}{\partial x_b \partial p_k}(x, p) \frac{\partial x_b}{\partial \alpha_j} \\ &= W_{\star ai} [D_x D_p \Phi(x, p)]_{abc} V_{\star bj} \delta_{ck} \\ \implies D_\alpha D_p \phi &= W_\star^\top D_x D_p \Phi(x, p) [V_\star, I_m]. \end{aligned}$$

(A.3e) follows analogously, and (A.3d), (A.3f) follow by reversing the order of differentiation. $\square$

B. Bounding $\sum_{j,k} \left(\sum_{v=1}^n y_v W_{\star vi} W_{\star vj} W_{\star vk} \right)^2$

To bound $Q_i := \sum_{j,k} \left(\sum_{v=1}^n y_v W_{\star vi} W_{\star vj} W_{\star vk} \right)^2$, we first expand the square and rearrange the sum as follows:

$$\begin{aligned} Q_i &= \sum_{j,k} \sum_{v,\rho} y_v y_\rho W_{\star vi} W_{\star vj} W_{\star vk} W_{\star \rho i} W_{\star \rho j} W_{\star \rho k} \\ &= \sum_{v,\rho} y_v y_\rho W_{\star vi} W_{\star \rho i} \left(\sum_j W_{\star vj} W_{\star \rho j} \right) \left(\sum_k W_{\star vk} W_{\star \rho k} \right) \\ &= \sum_{v,\rho} y_v y_\rho W_{\star vi} W_{\star \rho i} (W_\star W_\star^\top)_{v\rho}^2. \end{aligned}$$

Using $W_\star W_\star^\top = I - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^\top \implies (W_\star W_\star^\top)_{v\rho} = \delta_{v\rho} - \frac{1}{n}$,

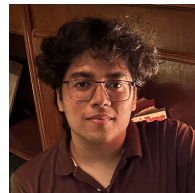
$$\begin{aligned} Q_i &= \sum_{v,\rho} y_v y_\rho W_{\star vi} W_{\star \rho i} \left(\delta_{v\rho} - \frac{1}{n} \right)^2 \\ &= \left(1 - \frac{2}{n} \right) \sum_{v=1}^n y_v^2 W_{\star vi}^2 + \frac{1}{n^2} \left(\sum_{v=1}^n y_v W_{\star vi} \right)^2. \end{aligned}$$

Using $\sum_v W_{\star vi}^2 = 1$ together with $|y_v| \leq m_2$ and applying the Cauchy–Schwarz inequality yields the desired result

$$\begin{aligned} Q_i &\leq m_2^2 \left(1 - \frac{2}{n} \right) + \frac{1}{n^2} \left(\sum_{v=1}^n y_v^2 \right) \left(\sum_{v=1}^n W_{\star vi}^2 \right) \\ &\leq m_2^2 \left(1 - \frac{2}{n} \right) + \frac{nm_2^2}{n^2} = m_2^2 \left(1 - \frac{1}{n} \right). \end{aligned}$$

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